

Double-side 2D/3D heterojunctions for inverted perovskite solar cells

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Defects at the top and bottom interfaces of three-dimensional (3D) perovskite photoabsorbers diminish the performance and operational stability of perovskite solar cells owing to charge recombination, ion migration and electric-field inhomogeneities^{1–5}. Here we demonstrate that long alkyl amine ligands can generate near-phase-pure 2D perovskites at the top and bottom 3D perovskite interfaces and effectively resolve these issues. At the rear-contact side, we find that the alkyl amine ligand strengthens the interactions with the substrate through acid–base reactions with the phosphonic acid group from the organic hole-transporting self-assembled monolayer molecule, thus regulating the 2D perovskite formation. With this, inverted perovskite solar cells with double-side 2D/3D heterojunctions achieved a power conversion efficiency of 25.6% (certified 25.0%), retaining 95% of their initial power conversion efficiency after 1,000 h of 1-sun illumination at 85 °C in air.

Engineering of perovskite interfaces is essential to enhance the performance and stability of perovskite solar cells (PSCs). 2D/3D perovskite heterojunctions have shown particular promise towards this end^{6–16}. Such heterojunctions are usually formed by post-treating an as-deposited 3D perovskite film with a solution that contains 2D ligands to reconstruct the 3D perovskite surface through cation exchange into a 2D perovskite^{7–9,12,14}. However, this strategy is applicable only at the top surface of 3D perovskite films. Moreover, it usually results in a 2D perovskite consisting of a mixture of multiple dimensionalities such as $n = 1, 2, 3$ and so on with random crystal orientations, in which n refers to the number of subsequent corner-sharing octahedral $[\text{PbI}_6]^{4-}$ inorganic slabs^{8,12,14}. Unfortunately, such mixed dimensionality may induce interfacial energetic inhomogeneity and could impede charge transfer through band misalignment when such a 2D/3D heterojunction is integrated into a charge-selective contact stack^{7,8,12,14}. By contrast, phase-pure 2D perovskite passivation can substantially reduce the charge-trap density and ion migration, leading to markedly improved device performance and stability^{10,11}. Hence, it is imperative to narrow the distribution of the different n layers in 2D perovskites^{10,11,16}. Yet, achieving phase-pure 2D perovskite passivating contacts with a proper crystal orientation is experimentally challenging^{10,11,16}.

Integrating 2D/3D heterojunctions at the buried bottom interface is desirable too, but has rarely been reported so far^{17–19}. The primary difficulty here is to avoid the dissolution of the pre-deposited 2D layer or the organic ammonium-based ligands during the subsequent 3D perovskite deposition^{17–19}. Indeed, most organic ammonium ligands are readily dissolved into the highly polar aprotic solvents typically used in 3D perovskite precursor inks such as dimethylformamide (DMF) and

dimethylsulfoxide (DMSO)^{11,20,21}. Hence, one approach for minimizing the dissolution of 2D ligands during perovskite solution casting is to strengthen their interaction with the substrate. This allows for immobilizing 2D ligands before perovskite deposition, and forming a 2D perovskite underneath the 3D perovskite film through cation exchange and intercalation of the 2D ligands into the perovskite film, post deposition.

2D/3D heterojunction at the bottom interface

We developed inverted (p–i–n) PSCs with a hole-collecting contact (p type) that is formed by anchoring 2PACz, a self-assembled monolayer (SAM), onto the transparent bottom electrode, consisting of indium tin oxide (ITO). To resolve the challenge of 2D/3D heterojunction formation at the bottom of the perovskite film, we mixed the 4-hydroxybenzylamine (HBZA) ligand into the 2PACz SAM solution, which was then coated onto the ITO bottom electrode. Here we reasoned that the amine head of the HBZA molecule would react with the phosphonic acid ($-\text{PO}(\text{OH})_2$) group of the 2PACz, forming an ionic bond, resilient against subsequent processing. This acid–base reaction is favourable with the lower values of the negative log of the acid dissociation constant for the first proton ($\text{p}K_{\text{a}1}$) of the phosphonic acid (about 2.5) and the negative log of the base dissociation constant ($\text{p}K_{\text{b}}$) of alkyl amines (about 3.6)^{22,23}. We note that only the first proton transfer is favourable between 2PACz and HBZA, owing to a higher $\text{p}K_{\text{a}2}$ value of the second proton (about 8.5)²². We experimentally verified that the above-mentioned reaction takes place in the mixture of 2PACz and HBZA, as shown in Extended Data Fig. 1a,b. Moreover, the $-\text{OH}$ group from HBZA can form hydrogen

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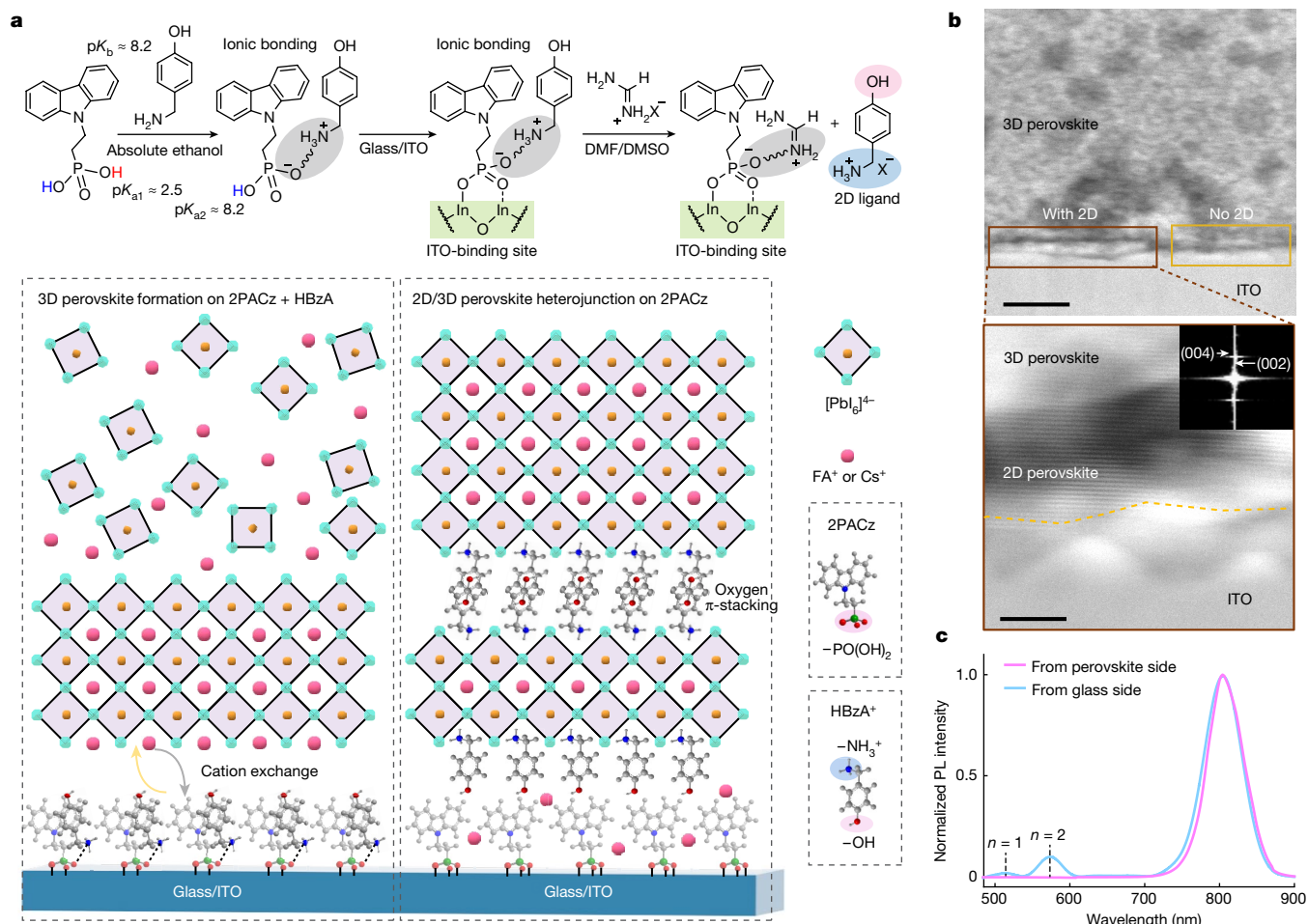


Fig. 1 | A 2D/3D heterojunction at the hole-selective bottom interface.

a, Top: the mechanism of 2D ligand formation and release from the ITO/2PACz surface. Bottom: the mechanism of 3D perovskite formation on ITO/(2PACz + HBZA) and cation exchange at the bottom interface (left); a 2D/3D heterojunction on the ITO/2PACz bottom contact (right). **b**, A cross-sectional high-angle annular dark-field STEM image of the near-bottom hole transport

layer contact, with a zoom-in of the area showing the parallel 2D perovskite layers. The fast Fourier transform shows 2D inter-planar spots (002) and (004) are matched with the dominant phase of $n = 2$. Scale bars, 50 nm (top) and 10 nm (bottom). **c**, Normalized PL spectra of the bottom-side 2D/3D from low to high wavelengths with different excitation directions from the glass side and top perovskite surface film using 405-nm excitation.

bonds with $-\text{PO}(\text{OH})_2$ and ITO, thus strengthening the attachment further²⁴. A detailed explanation of all these mechanisms is shown in Fig. 1a.

We then investigated the chemical properties of HBZA and 2PACz molecules on ITO using X-ray photoelectron spectroscopy. The high-resolution spectrum of the carbon region (C1s) of HBZA shows the presence of C=C, C-N and C-OH bonds that correspond to the aromatic ring, the amine head and the phenol group of HBZA, respectively (Extended Data Fig. 1c). Further, C1s spectra of 2PACz feature the signature of C-P bonds, corresponding to its $-\text{PO}(\text{OH})_2$ group. These identical features of 2PACz and HBZA were also found after their mixture was coated on ITO, indicating the presence of both molecules. To strengthen our hypothesis, we also carried out a washing test on the ITO/HBZA and ITO/(HBZA + 2PACz) samples with a highly polar aprotic DMF/DMSO solvent mixture that can easily solvate HBZA molecules by forming strong hydrogen bonding. We observed that if only HBZA layers are anchored on ITO, they could be easily washed away. However, the presence of 2PACz substantially enhances the attachment of HBZA onto the ITO surface, as indicated by a qualitative analysis of the marginal reduction of the atomic ratio of nitrogen to indium (N/In) after washing (Supplementary Table 1). This supports our hypothesis that HBZA and 2PACz form an ionic bond through proton transfer from phosphonic acid to the amine group. This bonding

facilitates their adherence to the substrate during perovskite processing and assists in the formation of a 2D perovskite layer beneath the 3D perovskite layer.

After successfully attaching HBZA ligands on the ITO/2PACz surface, we deposited the 3D perovskite ink through spin-coating. The HBZA ligands are liberated after 3D perovskite deposition, initiating the 2D perovskite formation at the bottom interface through cation exchange between HBZA⁺ and formamidinium (FA⁺) or caesium (Cs⁺) of the 3D perovskite ink. This cation exchange process is completed during the thermal annealing of perovskite films at 120 °C for 30 min to crystallize the 2D/3D perovskite film. To prove this mechanism, we completely removed the resulting perovskite layers from the ITO/2PACz/2D-HBZA/3D-perovskite structure by washing the samples with a DMF/DMSO solution and measured the work function of the resulting structure using kelvin probe force microscopy. We noted that the work function of the bare ITO/(2PACz + HBZA) sample remained similar before and after washing with DMF/DMSO (5.08 eV; Extended Data Fig. 2a–c). However, after the perovskite film deposition and subsequent removal of the HBZA ligands, the work function of ITO/(2PACz + HBZA) is shifted to 5.42 eV, closer to the work function of pristine ITO/2PACz (5.40 eV) with identical conditions (Extended Data Fig. 2d,e). This indicates that DMF/DMSO alone was not sufficient to remove HBZA molecules. Rather, the removal was facilitated by the 3D

perovskite deposition, corroborating our hypothesis that 2D ligands are released only after perovskite film formation.

The cross-sectional high-resolution scanning transmission electron microscopy (STEM) images in Fig. 1b reveal the formation of a discontinuous 2D perovskite layer at the bottom interface between the ITO/2PACz and 3D perovskite layers. The discontinuity of the 2D perovskite layers might be related to their complex formation mechanism, such as a random distribution of HBzA molecules and their release during cation exchange. We further analysed the 2D/3D perovskite heterojunction at the bottom-side interface by photoluminescence (PL) spectroscopy and observed distinct PL emission peaks for 2D perovskite at about 570 nm and about 520 nm corresponding to $n = 2$ and $n = 1$, respectively¹² (Fig. 1c). The very high PL emission intensity of $n = 2$ indicates the dominant formation of near-phase-pure 2D perovskite layers. In addition, we carried out PL mapping at about 570 nm with different concentrations of HBzA ligand and observed discontinuities for 2D perovskite crystal PL emission at the bottom interface (Supplementary Fig. 1), in accordance with our high-resolution STEM results.

We further analysed the structural properties and chemical composition of 2D perovskite at the bottom-side interface by gently separating the bottom-side 2D/3D perovskite film from the ITO/hole-selective surface using the lift-off technique following a previous report²⁵. This approach preserves the chemical and structural integrity of the film (see the schematics of the process in Supplementary Fig. 2a). We then characterized the bottom-side perovskite film by scanning electron microscopy and X-ray photoelectron spectroscopy measurements. Scanning electron microscopy top-view images of the lift-off samples show no distinctive morphological damage to the control and 2D/3D heterojunction samples (Supplementary Fig. 2b). However, we note that the crystal grain sizes of perovskite of 2D/3D samples are larger ($1.64 \pm 0.31 \mu\text{m}$) than those of control perovskite ($0.93 \pm 0.20 \mu\text{m}$), indicating that 2D perovskite formation at the bottom interface also regulates the crystallization of the overlying 3D perovskite film²⁶. Further, the X-ray photoelectron spectroscopy analysis of 2D/3D samples revealed the presence of C–O and C–N bonds at 286.5 and 285.6 eV, associated with HBzA ligands (Supplementary Fig. 2c).

3D/2D heterojunction at the top interface

For the 3D/2D heterojunction at the top interface, we developed a simple two-step hybrid method to form phase-pure 2D perovskite layers. This process initially involves depositing the PbI_2 layer with controlled thickness through vacuum evaporation on top of the as-prepared 3D perovskite film (Fig. 2a, step (i)). This is then followed by dropping the HBzA salt or HBzA salt + formamidinium iodide solution on top of the perovskite/ PbI_2 film in step (ii). Finally, the samples are thermally annealed to form a 2D perovskite layer with controllable dimensionality and phase purity in step (iii).

Figure 2b shows the normalized PL spectra of 3D/2D perovskite films for two conditions. First, when coated with the solution containing only the HBzA salt cation, the 3D/2D perovskite sample shows a single emission peak at a low wavelength of about 520 nm (corresponding to $n = 1$)^{8,12,27}. Second, the sample based on the mixture of HBzA salt + formamidinium iodide also shows a single emission peak but at a higher wavelength (about 570 nm), which corresponds to $n = 2$ (refs. 8,12). These results confirm the success of our method to control the phase purity and dimensionality of 2D perovskites, which so far remained a challenge^{7,8,12,14}. By contrast, the conventional method of directly coating the HBzA salt solution on top of the 3D perovskite following only steps (ii) and (iii) (with the absence of PbI_2 deposition) forms a mixture of multiple dimensionalities of 2D perovskite ($n = 1, 2$ and 3; Supplementary Fig. 3), consistent with previous works^{8,12}. We note that a higher dimensionality of the 2D perovskite ($n > 1$) passivation layer is critical, owing to a higher conductivity by having less bulky organic cation interfaces^{8,11,12}.

Next we identified the crystal orientation and dimensionality of the 2D perovskite layers through grazing-incidence wide-angle X-ray scattering measurements. Figure 2c shows the diffraction map of grazing-incidence wide-angle X-ray scattering of the 3D/2D sample with strong diffraction peaks at lower q_z ($0.31 \text{ \AA}^{-1}$ and $0.64 \text{ \AA}^{-1}$) representing the (002) and (004) planes of the 2D perovskite ($n = 2$), respectively, with a layer spacing d of about $20.3 \text{ \AA}$ and about $9.8 \text{ \AA}$ (refs. 13,14,28,29). Strong and more discrete Bragg spots at a higher q_z (about $1.5 \text{ \AA}^{-1}$ and about $1.7 \text{ \AA}^{-1}$) were observed in the 3D/2D sample. This indicates the improved orientation of the 2D perovskite corresponding to the (282), (121) and (082) planes, formed with perpendicular orientation to the 3D perovskite underneath, as highlighted with oval dashed outlines^{30,31}. By contrast, we do not observe the diffraction peaks at lower q_z and discrete Bragg spots in control perovskite samples (Supplementary Fig. 4a,b). This result is also consistent with the X-ray diffraction pattern of the phase-pure 2D perovskite thin film, as shown in Supplementary Fig. 4c,d. Further, we visualize the 2D perovskite formation and orientation at the top contact using cross-sectional high-resolution STEM in Fig. 2d, with a predominantly perpendicular orientation. In addition, a thin layer of 2D perovskite does not change the morphology of the 3D perovskite underneath (Supplementary Fig. 5), suggesting the robustness of our hybrid method to form perovskite heterostructures.

Device performance and characterization

After successfully completing the near-phase-pure 2D perovskite passivation layers at the bottom and top 3D perovskite interfaces, we applied them to PSCs with a structure of ITO/2PACz/2D-perovskite e/3D-perovskite/2D-perovskite/ C_{60} /bathocuproine/Ag (Fig. 3a). We first tested various alkyl amine-based 2D ligands with different concentrations with respect to the ratio between the 2PACz SAM and the 2D ligands at the bottom-side 2D/3D passivation, as shown from the statistical analysis of device performance in Extended Data Fig. 3a,b. The device results revealed that the optimum concentration of the 2D ligand at the bottom contact is around a 1:1 molar ratio with 2PACz SAM. The higher concentration of the 2D ligand at the bottom interface leads to a thicker 2D perovskite layer, lowering the fill factor (FF) of the devices. This is probably due to the higher resistance of the long alkyl ligands in a parallel orientation to the 3D perovskite layer, which can inhibit charge transport³². Among the five alkyl amine ligands tested, the HBzA ligand gives the highest performance, potentially owing to the addition of an –OH group that could form a hydrogen bond with $-\text{PO}(\text{OH})_2$ and ITO or coordinate with 3D perovskite at the 2D/3D interface^{33,34}. The longer 2D ligand (that is, hexylmethylamine) results in a lower performance, potentially owing to a higher charge transfer resistivity induced by long alkyl ligands. This suggests the critical roles for additional functional groups and the need to design appropriately sized 2D ligands for bottom passivation^{8,16}.

Next we fabricated the top-side 3D/2D heterojunction using the above-mentioned hybrid method. Here we deposited PbI_2 with different thicknesses (7, 15 and 25 nm) and found that the thickness of the 2D perovskite depends on the initial thickness of the inorganic PbI_2 layer. On the basis of the device results, it is found that a 15-nm-thick PbI_2 film results in a 35-nm-thick 2D perovskite layer, which is optimal for top-side passivation (Extended Data Fig. 3c). By contrast, a 25-nm PbI_2 film forms a thicker 2D perovskite (around 55 nm; see the cross-sectional scanning electron micrograph in Supplementary Fig. 6), which slows down the charge transfer and extraction rate at the perovskite/ C_{60} interface. This is consistent with the decreased FF values from these devices, as indicated by the transient PL measurement and differential lifetime analysis in Supplementary Fig. 7.

We then investigated the relation between the 2D perovskite orientation and charge transfer and extraction rates. For this, we compared the charge transfer and extraction rates between the 3D perovskite and C_{60} when a 35-nm-thick 2D perovskite layer prepared by the hybrid

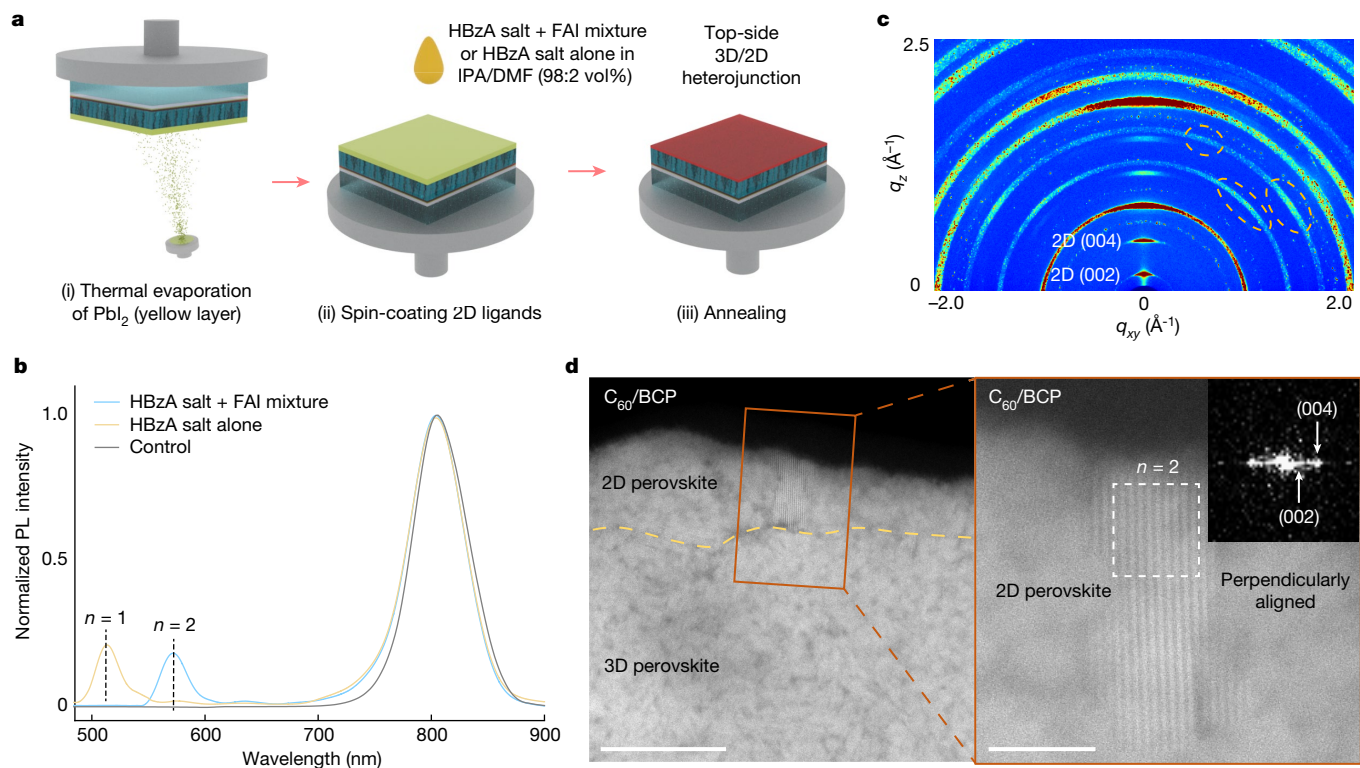


Fig. 2 | A 3D/2D heterojunction at the electron-selective top interface.

a, The two-step hybrid process to form 3D/2D heterojunction perovskites. FAI, formamidinium iodide. **b**, Normalized PL spectra of each film with the light direction from the top perovskite surface film. **c**, Grazing-incidence wide-angle X-ray scattering maps of the top-side 3D/2D heterojunction at a low incidence

angle (0.2°). **d**, Cross-sectional high-angle annular dark-field STEM image of the near-top C_{60} contact, with a zoom-in of the area showing the perpendicular orientation of the 2D perovskite. BCP, bathocuproine. Scale bars, 50 nm (left) and 10 nm (right).

and solution post-treatment methods is inserted between them. It should be noted that the 2D perovskite layer prepared with the solution post-treatment method has a parallel orientation to the 3D perovskite surface, which substantially hinders charge transport, and limits the thickness of the 2D layer^{8,11,12,16,35}.

The charge transfer and extraction rates at the interfaces also depend on the energetic alignment between the 3D perovskite and the charge-selective contacts (Extended Data Fig. 4a). With the two-step hybrid method, the secondary electron cutoff of the 2D perovskite shifted to a lower binding energy (Extended Data Fig. 4b). As a result, the conduction band minimum of the top-side phase-pure 2D perovskite ($n = 2$) prepared by the hybrid method was deeper compared to the previously reported mixed phase of 2D perovskite top-contact passivation using the conventional solution post-treatment method, which resulted in markedly minimizing the electron barrier^{8,12,16}. Further, the slightly deeper valence band maximum of the bottom-side 2D perovskite compared to that of the control 3D perovskite is favourable for hole transfer at the ITO/2PACz contact interface (Extended Data Fig. 4c,d).

We then adopted double-side 2D/3D heterojunction passivation to our inverted PSCs and demonstrated a maximum power conversion efficiency (PCE) of 25.63% (open-circuit voltage (V_{oc}) = 1.19 V, short-circuit current density (J_{sc}) = 24.94 mA cm^{-2} and FF = 85.9%; Fig. 3b) under reverse scan conditions and a stabilized PCE of 25.1% using the triple-cation $\text{Cs}_{0.025}\text{MA}_{0.075}\text{FA}_{0.90}\text{PbI}_3$ perovskite with a bandgap of 1.53 eV (Supplementary Fig. 8). This result represents an absolute PCE enhancement of about 2.3% compared to the PCE of control devices (around 23.3%) and >1% compared to those of only top- or only bottom-side 2D/3D heterojunctions with a PCE of around 24% (refs. 8,10–12). Figure 3c shows the statistical analysis results of the PCE, V_{oc} and FF values from 25 devices for each condition. We demonstrated the universality of our approach by testing various hole-transporting

SAM molecules (that is, 2PACz, MeO-2PACz and 4PADCB) and other perovskite compositions of $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$ (about 1.54 eV) with PCEs reaching up to 25% for each condition, as shown from the statistical analysis of devices in Supplementary Fig. 9 and Extended Data Fig. 5, respectively. We also confirmed the high process robustness of our approaches by showing less than 1% PCE deviation from person to person, carried out by three different researchers (Supplementary Fig. 10). The performance of the double-side 2D/3D heterojunction passivated solar cells was verified by an independently accredited testing centre by presenting a PCE of 25.00% (V_{oc} = 1.17 V, J_{sc} = 25.00 mA cm^{-2} and FF = 85.7% under a reverse current–voltage scan; Supplementary Fig. 11). This result represents one of the highest certified efficiencies in inverted PSCs in the current literature (Supplementary Table 2).

Notably, the $V_{oc} \times \text{FF}$ value of our double-side 2D/3D heterojunction-based device was around 91% with respect to the Shockley–Queisser limit, one of the highest values among PSCs with only top-side or bulk passivation (Supplementary Fig. 12). The high V_{oc} and FF values of the devices can be associated with the low charge recombination and traps at the top and bottom interfaces^{15,36}. First, we measured the PL quantum yield to demonstrate the quality of the perovskite film before and after 2D perovskite passivation. The substantially improved PL quantum yield of the double-side 2D/3D passivation ($6.67\% \pm 0.58\%$) confirms the high quality of the 2D/3D/2D perovskite film compared to the control perovskite ($1.84\% \pm 0.63\%$) and the one-side 2D/3D passivation (Extended Data Fig. 6a) samples. Even after adding the C_{60} layer, the double-side 2D/3D perovskite maintained a high PL quantum yield value ($5.78\% \pm 0.33\%$) compared to that of the control sample (substantially reduced to around $0.19\% \pm 0.06\%$), demonstrating the improved optoelectronic quality of the perovskite layer with the full stack of devices. This suggests that the 2D perovskite passivation layer can efficiently suppress the charge recombination at these interfaces.

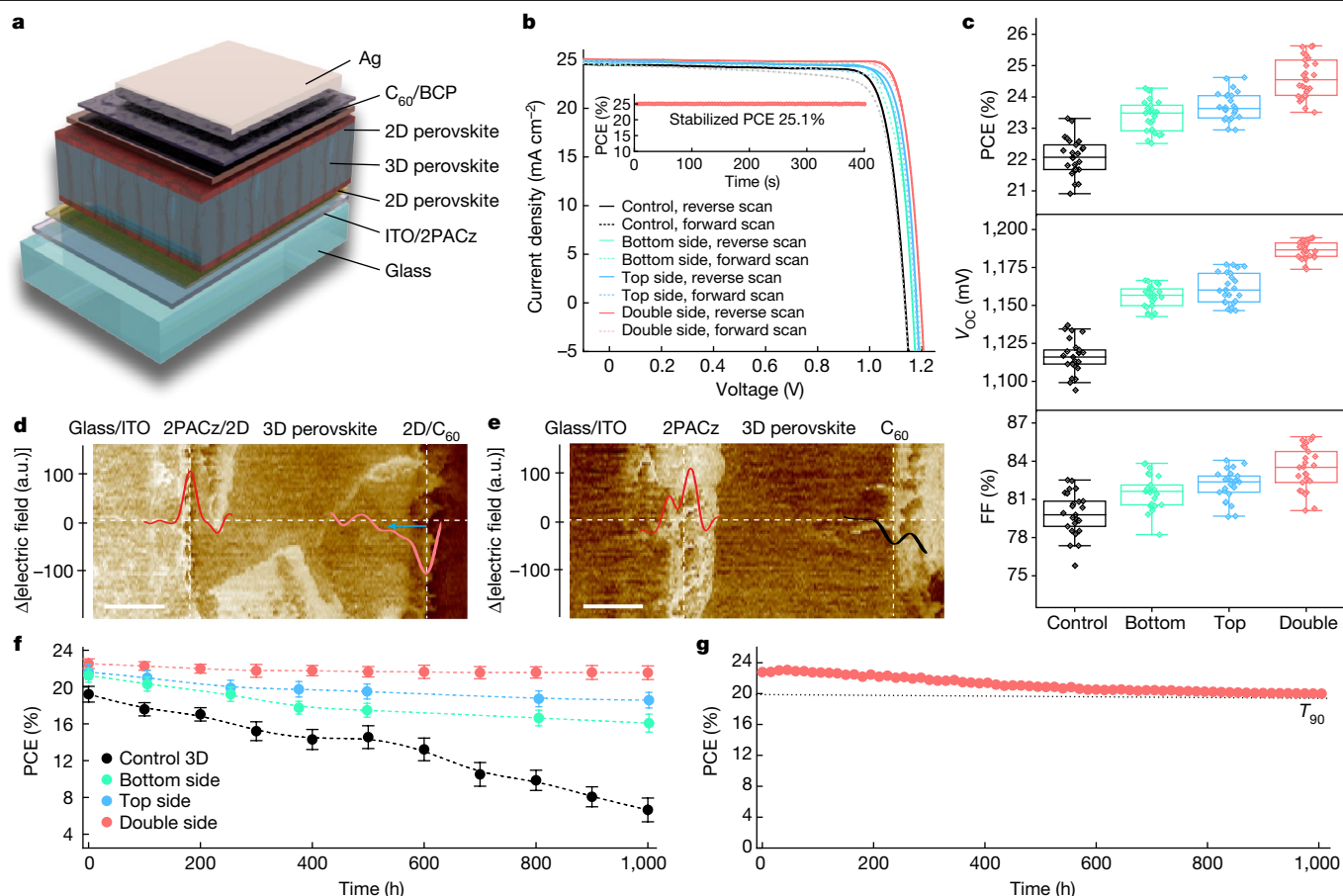


Fig. 3 | Device performance and stability analyses. **a**, A sketch of the fabricated single-junction PSC. **b**, Representative J - V curve for control, bottom-, top- and double-side 2D/3D heterojunction PSCs. **c**, Statistics parameters of PCE, V_{oc} and FF (from top to bottom) values of PSCs. The box and whisker plots indicate the statistical distribution: the centre line represents the average value; the bottom and top of the box represent the 25% to 75% percentiles; the small square represents the mean; the whiskers represent outliers. **d, e**, Electric field distribution through both interfaces from the C_{60} electron transport layer to the glass/ITO side acquired through Kelvin probe force microscopy cross-sectional scans. We mapped the photo-voltage of the cells illuminated by a light-emitting

diode. Note that we break the hole- and electron-selective peaks for better visualization. a.u., arbitrary units. Scale bars, 200 nm. **f**, Stability of encapsulated control bottom-, top- and double-side 2D/3D passivation-based devices (ITO/2PACz/CsFAPbI₃/C₆₀/SnO₂/IZO/Cu) under 1-sun illumination at 85 °C and open-circuit conditions. The data represent the mean standard deviation for five cells. **g**, Maximum power point tracking of an encapsulated double-side 2D/3D passivation-based device (ITO/2PACz/2D/3D-CsFAPbI₃/2D/C₆₀/SnO₂/IZO/Cu) under 1-sun illumination at 40 °C. T_{90} represents the lifetime of the cell with retention of 90% of their initial PCE value.

These results are also consistent with the longer Shockley–Read–Hall lifetime of double-side 2D/3D heterojunctions (2.69 μ s), around three and two times longer versus the control (0.97 μ s) and only one-side passivation (1.34 μ s and 1.73 μ s for the only bottom- and top-side passivation, respectively), as shown in Extended Data Fig. 6b.

We also determined the trap states for control and double-side 2D/3D passivation of perovskite films using space-charge-limited current and thermal admittance spectroscopy analyses^{12,37}. For the space-charge-limited current analysis, we fabricated electron-only (ITO/SnO₂/perovskite/C₆₀/bathocuproine/Ag) and hole-only (ITO/SAM/perovskite/PTAA/MoO_x/Ag) devices to estimate the trap densities of perovskite films. The trap-filled limited voltage (V_{TFL}) of the double-side 2D/3D passivation samples was more than twofold lower than that of control samples for both electron- and hole-only devices (Extended Data Fig. 6c,d), indicating reduced trap density states (N_t) for both electron- and hole-only devices (Supplementary Note 1). This result is indicative of reduced levels of negative and positive defects at both interfaces with 2D perovskite passivation. It should be noted that the space-charge-limited current analysis has limitations in extracting the exact value of N_t owing to the presence of mobile ions in perovskite materials³⁸. Therefore, we also carried out thermal admittance spectroscopy analysis to quantify the N_t of perovskite films relative

to their energetic defect levels^{12,37}. Overall, the N_t of double-side 2D/3D perovskite passivation-based devices has lower N_t values in all energetic defect levels, suggesting sufficient passivation of shallow and deep levels of defect in the perovskite film by 2D perovskite passivation (Extended Data Fig. 6e).

Further, the defects at the interfaces of perovskite films can affect the charge accumulation and electric-field inhomogeneity of perovskite devices^{1,3–5,37}. We then analysed the capacitance–voltage under various light illumination intensities to evaluate the charge dynamics at the interfaces (Supplementary Fig. 13). The shift in peak potential (V_{peak}) under reverse bias with respect to the illumination intensity reflected the degree of interfacial charge accumulation, which was due to interfacial charge trapping³⁷. Marginal shifting of the V_{peak} in the double-side 2D/3D device indicated a reduced charge accumulation (less charge trapping) compared to that of the control device (larger V_{peak} shifting), consistent with the above results. From the cross-sectional Kelvin probe force microscopy mapping (Fig. 3d,e), stronger electric-field differences after 2D perovskite insertion at the perovskite/C₆₀ contact further prove a lower interfacial defect density³⁹. We note that the electric field extended further into the perovskite film (see the blue arrow), more effectively repelling holes and thus suppressing recombination at the perovskite/C₆₀ interface⁴⁰. We note that a similar electric-field

strength difference is observed at the 2PACz/perovskite contact for both control and double-side passivation devices, presumably due to the higher dipole of 2PACz SAMs³⁹. However, a uniform electric field is observed after 2D perovskite insertion at the 2PACz/perovskite contact, as indicated by a single peak. Consequently, this aids in charge transfer and extraction and leads to enhance V_{oc} and FF values of devices.

Stability and ion migration analyses

We evaluated the long-term stability of encapsulated PSCs under light and thermal stresses. First, Fig. 3f shows the stability under the open-circuit condition at 1-sun and 85 °C of the devices in air (with a relative humidity of about 50%). The double-side 2D/3D heterojunction devices delivered a relative PCE loss of only 5% after more than 1,000 h. By contrast, the bottom- and top-side passivation-based devices retain around 75% and 86% of their initial PCEs, respectively, versus the control devices (without any 2D/3D heterojunction), which retain around only 35% under identical conditions. This result indicates the importance of passivating perovskite film's top and bottom interfaces to improve their stability. Further, we measured the operational stability of double-side 2D/3D heterojunction devices under maximum-power-point tracking. Around 90% of their original PCE was retained after the PCE tracking for over 1,000 h under 1-sun illumination at 40 °C (Fig. 3g).

The degradation of perovskites under illumination and thermal stress can be related to defect-induced ion migration^{1,25,37,41}. First, the defects related to iodine generation and migration were investigated under 1-sun illumination at 85 °C. It has been reported that the substantial release of iodine compound from perovskite film during light soaking at elevated temperature is mainly due to interstitial iodide or PbI_2 defects^{25,41}. Passivating those defects by double-side 2D/3D passivation suppresses the iodine generation and migration in perovskite films. This is proved by the lowering intensity of the UV-Vis absorption spectrum of iodine at the maximum peak of wavelength around 500 nm after immersing the perovskite films in toluene solution and exposing them to light and heat^{25,41}, (Extended Data Fig. 7a,b).

Finally, the activation energy for ion migration was obtained to reveal the ion-induced migration character in 2D/3D and control 3D perovskite films using temperature-dependent conductivity measurements³⁷. The activation energy value of the control 3D perovskite was fitted to be 0.27 eV, in agreement with a previously reported value. By contrast, the activation energy value for the double-side 2D/3D heterojunction film is increased to 0.47 eV (Extended Data Fig. 7c). This result indicates that the double-side 2D/3D heterojunctions have a substantially boosted energy barrier for ion migration, which might also improve the perovskite crystal stability.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-07189-3>.

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Methods

Materials

DMF (99.8%, anhydrous), DMSO (99.9%, anhydrous), potassium chloride (KCl, 99.9%), HBzA, (C_7H_5NO), benzylamine, phenethylamine, butylamine, hexylmethylamine and caesium iodide (CsI, 99.9%) were all purchased from Sigma Aldrich. FAI and methylammonium chloride (MACl) were purchased from Xian-Polymer. Lead iodide (PbI_2 , ultradry) was purchased from Alfa Aesar. 4-Hydroxybenzylammonium salts were purchased from Alfa Chemical. 2PACz ([2-(9H-carbazol-9-yl)ethyl] phosphonic acid, >98.0%) and MeO-2PACz were purchased from TCI. 4PADCB (4-(7H-dibenzo[c,g]carbazol-7-yl)butyl)phosphonic acid) was purchased from Dyenamo. C_{60} (>99.5% purity) and BCP (>99%) were purchased from Ossila.

Device fabrication

Glass/ITO substrates were washed in acetone and then isopropyl alcohol (20 min in each). After washing, the substrates were dried and treated with UV ozone before use for 10–20 min. The hole-transporting SAM (1–2 M) or mixed SAMs + 2D ligands (1:0.5, 1:1, 1:1.5 and 1:2 molar ratio) were dissolved in ethanol. Then the solution was spin-coated onto glass/ITO at 5,000 r.p.m. for 30 s in a nitrogen-filled glovebox. After thermal annealing at 100 °C for 10 min, the hole-selective-layer-coated ITO was cooled down and washed with ethanol by dynamic coating at 5,000 r.p.m. for 30 s to remove unbound molecules.

A perovskite precursor solution (about 1.5 M) of $Cs_{0.025}MA_{0.075}FA_{0.90}PbI_3$ was prepared by mixing FAI, MAI, CsI and PbI_2 with a molar ratio of 0.90:0.075:0.025:1 and then dissolving them in a mixed solvent of DMF and DMSO (4:1 volume ratio). A 15 mol% amount of MACl was also added. $Cs_{0.05}FA_{0.95}PbI_3$ was prepared by mixing FAI, CsI and PbI_2 with a molar ratio of 0.95:0.05:1 and then dissolving them in a mixed solvent of DMF and DMSO (4:1 volume ratio) to get about 1.5 M. Then, around 10 mol% of MACl and 2 mol% of KCl were also added into the mixed perovskite solution. The mixture of precursor solutions was shaken by a vortex shaker overnight at room temperature until fully dissolved.

The perovskite precursor solution was spin-coated at 2,000 r.p.m. for 40 s and 6,000 r.p.m. for 10 s. Immediately before the end spin-coating process, 200 ml of anisole was dropped onto the films. Then the films were annealed at 120 °C for 30 min to form the perovskite layer. For 2D perovskite passivation at the top side, at the first step, PbI_2 beads were thermally evaporated (10^{-6} torr) at an evaporation rate of 0.5 Å s^{-1} to get a thickness of around 7, 15 and 25 nm. The second step involved dropping HBzA salt solution or a mixture of HBzA salt and FAI solution with a 1:1 molar ratio (in a mixture of IPA and DMF, 98:2 vol%) and then spin-coating at 2,000 r.p.m. for 30 s, followed by thermal annealing at 100 °C for 10 min. After this process, the perovskite films were stored overnight in vacuum conditions. The whole process was carried out in a N_2 glovebox with O_2 level <10 ppm and H_2O level <0.1 ppm. Then the samples were transferred to a thermal evaporator for the C_{60} (25 nm) and BCP (3 nm) deposition. For the final step, a 120-nm-thick Ag layer was evaporated at low pressure ($<10^{-6}$ torr) with an area of about 0.1 cm^2 . An approximately 120-nm layer of MgF_2 was also evaporated at the glass side to minimize the reflection losses from the glass substrates.

Device analysis

The $J-V$ curves were recorded using a Keithley 2400 source unit under AM 1.5 G (100 mW cm^{-2}) illumination with an Abet Technologies Sun 3000 solar simulator. The spectral mismatch was calibrated using a KG-5 filter-covered mono-silicon standard cell (Newport). All devices were measured in nitrogen-filled glovebox for a sweep mode of reverse and forward scans with a scanning rate of 0.1 V s^{-1} . The stabilized output of the devices was acquired by recording the power output of the illuminated device at a constant voltage near the maximum power point extracted from the $J-V$ curve. External quantum efficiency spectra were obtained by passing the output of a 400-W xenon lamp through

a monochromator and filter. Calibration was carried out with a 603621 calibrated silicon and germanium reference detector. For activation energy measurement, the electrical poling was carried out by applying different biases to the devices for 1 min in the dark; current–voltage was measured with a Keithley 2401 source. Temperature dependence was measured with a CTI cryogenic probe station with an attached Lakeshore 331 temperature controller under vacuum (1×10^{-4} Pa). The temperature was monitored using a thermocouple placed in contact with the sample. During the measurement, devices were first cooled to 10 K for 10 min and then heated to adjust the temperature. The temperature was stabilized for 5 min before recording the current for each step. Cross-sectional Kelvin probe force microscopy measurements were carried out using a home-built system based on a Bruker Dimension Icon atomic force microscope. The sample preparation and measurements were conducted inside an Ar-filled glovebox. We used a Pt–Ir-coated silicon probe (PPP-EFM) and operated in tapping mode. The mapping was carried out with 1,024 pixels in the fast-scan axis and more than 32 lines in the slow-scan axis, at a scan rate of 0.35 Hz. To ensure a flat surface and expose all layers of the device, the cell was directly cleaved without polishing or ion-milling. We then mapped the surface potential at the same location under dark and LED illumination, with the LED providing white light of approximately 1-sun intensity. To reduce the noise level, we averaged the electrical potential profiles from all slow-scan lines. Next we subtracted the dark potential from the illuminated condition and calculated the first derivative to obtain the electric field distribution across the device.

Sample characterization

Field-emission scanning electron micrographs were obtained with a JSM-7610F device (JEOL). The PL emission was spectrally resolved using a collection optic. For the STEM-based study, a cross-sectional electron-transparent lamella was prepared in a focused ion beam-equipped SEM-FIB Helios G4 DualBeam, FEI with the help of an EasyLift nanomanipulator and a Ga ion source. First, to protect the region of interest during focused ion beam processing, two types of protective coating were deposited: a $0.5\text{-}\mu\text{m}$ layer of Pt deposited by the electron beam, followed by a $3\text{-}\mu\text{m}$ layer of Pt coating deposited by the ion beam for final protection. A stepwise beam milling procedure with decreasing beam current (2.4, 0.44, 0.26, 0.045, 0.025 nA for 30 to 5 kV) was carried out by cutting and thinning down the lamella to 60 nm, while decreasing the beam current (until 0.025 nA at 2 kV) to avoid ion beam damage. Additionally, a low-current cleaning process (5–2 kV, 81–28 pA) was carried out to remove any potential contamination. STEM-based experiments such as high-angle annular dark-field STEM were carried out in a Cs probe-corrected ThermoFisher Titan 60-300 Cubed transmission electron microscope operating at 300 kV. Transmission electron microscopy data processing was carried out using Gatan Digital Micrograph and Thermo Scientific Velox suites. For transmission electron microscopy and PL characterization, the perovskite film sample preparation used a 1:2 molar ratio of SAM and 2D ligand to help observation of 2D perovskites.

Grazing-incidence wide-angle X-ray scattering analysis was conducted in the PLS-II 6D UNIST-PAL beamline of the Pohang Accelerator Laboratory, Korea. The energy of the X-rays from the bending magnet was monochromatized to 18.986 keV (0.6530 Å) using a double-crystal monochromator. The 2D charge-coupled device detector (MX225-HS, Rayonix) was used to record the 2D grazing-incidence wide-angle X-ray scattering pattern, and the grazing-incidence wide-angle X-ray scattering sample chamber was equipped with a five-axis motorized stage for fine sample alignment. The sample-to-detector distance was 242.58 mm, and the diffraction angles were calibrated using NIST SRM660b, lanthanum hexaboride (LaB_6). An incidence angle of about 0.2° was used for the grazing-incidence wide-angle X-ray scattering measurements to obtain crystallographic information on the perovskite films used in this work.

X-ray photoelectron spectroscopy and ultraviolet photoelectron spectroscopy measurements were carried out in an ultrahigh vacuum chamber (ScientaOmicron) operating at a pressure of 5×10^{-10} mbar. Ultraviolet photoelectron spectroscopy measurements were carried out using a 21.2-eV vacuum UV source (focus). The sample was biased by 10 eV to observe the low kinetic energy cutoff. The photoelectrons were collected at an angle of 80° between the sample and analyser, with a normal electron take-off angle. The constant analyser pass energy was 5 eV for the valence band region and for the secondary electron cutoff. X-ray photoelectron spectroscopy was carried out in the same spectrometer, equipped with a monochromatic Al K α X-ray Omicron XM1000 X-ray source ($h\nu = 1,486.6$ eV) operating at a power of 390 W. The high-resolution spectra were collected at a constant analyser energy of 15 eV. The spectra were analysed with Casa XPS software. The individual peak envelopes were fitted by a Gaussian–Lorentzian (GL30) using a Tougaard-based background function. Fourier-transform infrared spectroscopy (FTIR) was conducted using a Cary 600 Series FTIR Spectrometer (Agilent Tech.) equipped with PIKE GladiATR geometry using a diamond crystal in transmission mode. Samples of powder were first dissolved in ethanol solvent and dried in a vacuum, and then annealed at 100°C in a N_2 glovebox for 10 min to completely remove the solvent. The dried powder samples were spread uniformly above the crystal surface of the FTIR. The spectral resolution was set to 4 cm^{-1} and obtained by averaging 20 scans of each measurement. Transient PL decay and PL quantum yield were recorded using a spectrofluorometer (Fluoroma-Modular, Horiba Scientific). The 520-nm calibrated pulse laser was generated from a picosecond laser diode (PicoQuant) equipped with a 650-nm band-pass filter to collect 795-nm emitted light. The fluence is estimated to be around 10 nJ cm^{-2} . For PL quantum yield, an integrating sphere (Fluorolog, Horiba JobinYvon) was used and compared with different tools using a JASCO FP-8500 spectrofluorometer equipped with an ILF-835 integrating sphere with the same procedure. The results were consistent between two different setups and measurements, confirming the reliability of our results, as shown by the PL quantum yield data in Extended Data Fig. 6a.

Device stability testing

For the stability test, an approximately 10-nm layer of tin(IV) oxide and an 80-nm layer of indium zinc oxide were deposited on top of the C_{60} sequentially, by atomic layer and sputtering depositions, respectively, to replace the BCP layer. The operational stability tests were carried out at the maximum-power-point tracking for the encapsulated devices

under 1-sun illumination in ambient air (relative humidity $> 50\%$) at about 40°C . The voltage at the maximum-power-point tracking was automatically applied, and the power output of the devices was tracked. For the open-circuit light soaking test, the devices were kept under full 1-sun illumination and constant temperature at 85°C on a hot plate in ambient air (relative humidity $> 50\%$). The devices were then measured periodically until 1,000 h. The devices were encapsulated with a simple glass and epoxy method. All processes were carried out in a N_2 glovebox.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author upon request.

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Author contributions R.A. conceived the idea, directed the research, and wrote the original draft. R.A., D.S.U., I.F.I. and A.P. fabricated the solar cells. B.V. conducted focused ion beam and transmission electron microscopy measurements and analyses. A.M.R. carried out the grazing-incidence wide-angle X-ray scattering measurement and analysis. R.A. and A.M.R. carried out PL quantum yield measurement and analysis. S.Z. contributed to conceptualizing the idea of 2D perovskite at the bottom contact and related discussion. P.D. carried out X-ray photoelectron spectroscopy, inverse photoemission spectroscopy and ultraviolet photoelectron spectroscopy measurements and data analysis. R.A., D.S.U. and E.U. carried out optical spectroscopy measurements and data analysis. D.S.U. carried out scanning electron microscopy measurements. A.P. carried out the top-surface Kelvin probe force microscopy measurement and data analysis. R.A. carried out and analysed the electronic characterization of the devices. F.C. carried out cross-sectional Kelvin probe force microscopy and analysis. D.S.U. and A.A.S. carried out the FTIR and analysis. I.F.I. helped synthesize and characterize 2D perovskite layers. I.F.I. and A.A.S. carried out the X-ray diffraction measurement analysis. A.R.P. fabricated the aperture mask and determined the area. A.A.S. and A.S.S. helped review the manuscript. E.A. contributed to reviewing and writing the manuscript. S.I.S. and C.X. supervised the project. S.D.W. supervised the overall project and secured the funding. All authors contributed to the manuscript.

Competing interests The authors declare no competing interests.

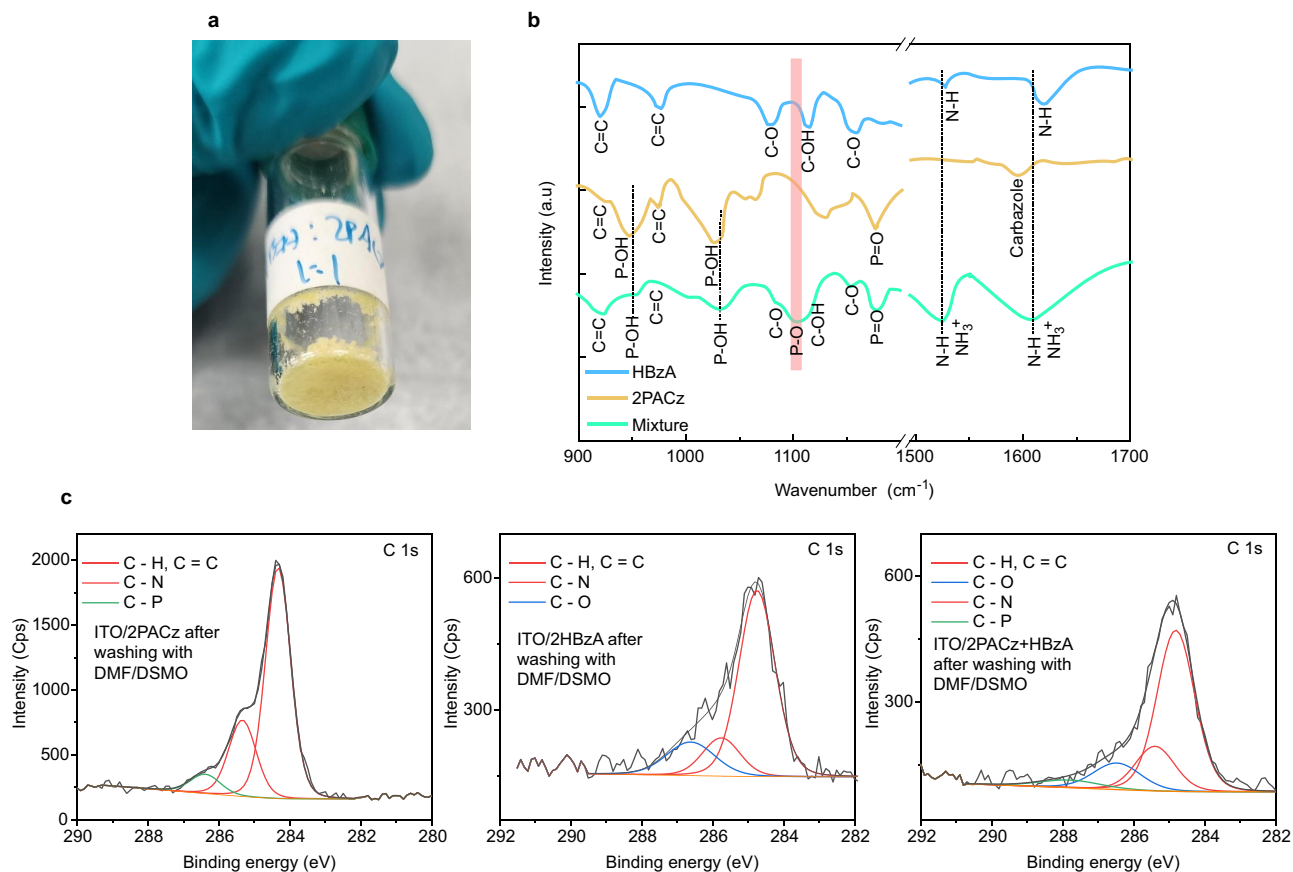
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-024-07189-3>.

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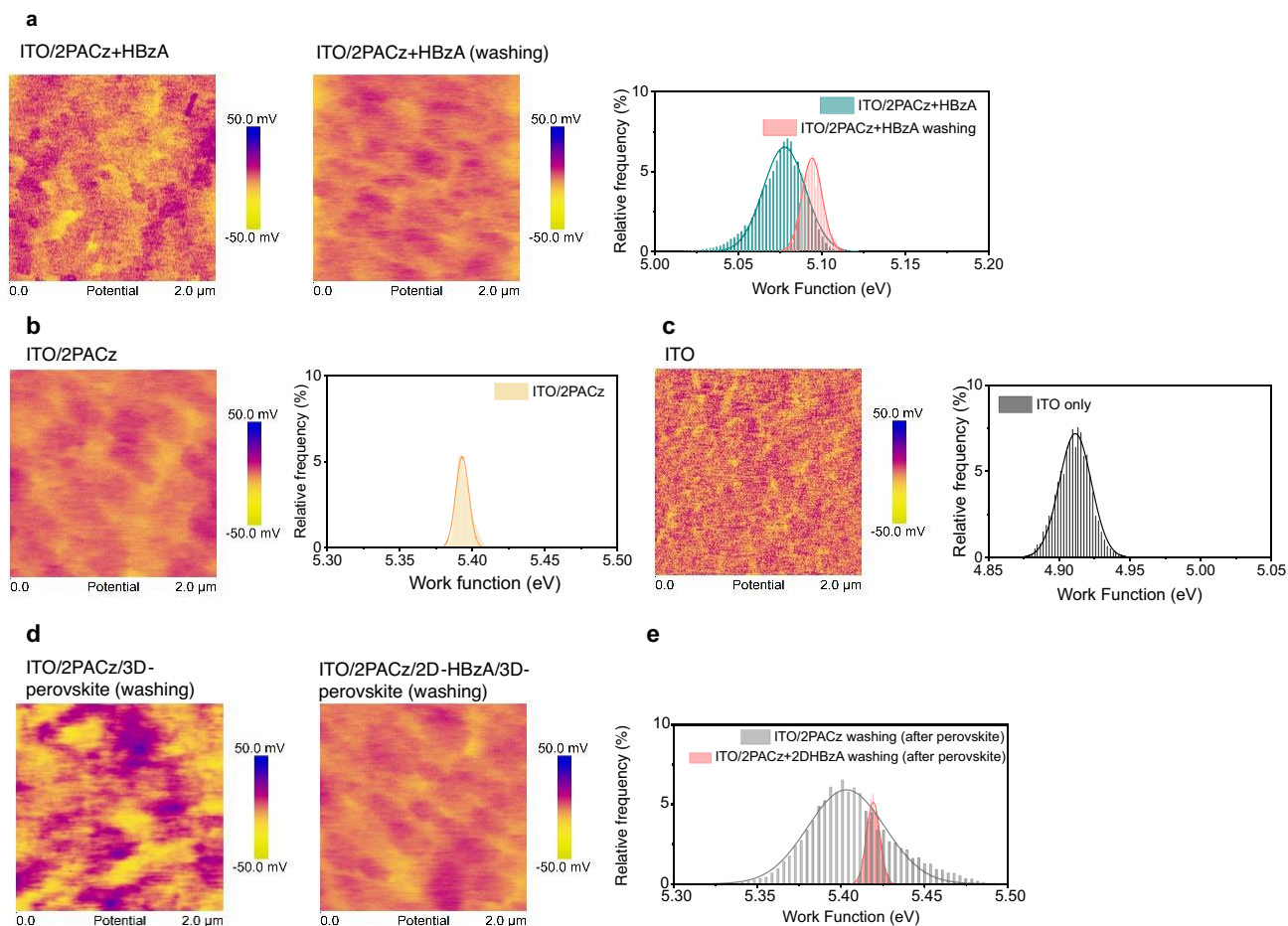
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Extended Data Fig. 1 | Chemical analysis of SAMs and HBzA ligand.

a, Photograph of the mixture 2PACz:HBzA (1:1 molar ratio) compound after drying under vacuum for a few days and then annealed at 100 °C in a nitrogen glovebox to completely remove the solvent. The yellow precipitated powder indicates the protonation between the first P–OH of the phosphonic group of 2PACz and the amine of HBzA molecules, forming an ionic bond. **b**, FTIR peaks of HBzA, 2PACz, and mixture (2PACz:HBzA, 1:1 molar ratio) in solid-state powder. The shifting peaks of P–OH at 950 cm^{-1} and 1033 cm^{-1} to the higher wavenumber of the mixture compound indicate stronger bonding due to deprotonation

of the first P–OH group of the phosphonic acid of 2PACz. Consequently, the additional new peak associated with the P–O bond at 1105 cm^{-1} is formed. In contrast, N–H bending peaks of the amine group (HBzA) at 1530 cm^{-1} and 1620 cm^{-1} are shifted to the lower wavenumber with enhanced intensity in the mixture compound, presumably due to the protonation of $-\text{NH}_2$ to $-\text{NH}_3^+$. These results successfully confirm the acid-base reaction between the pK_{a1} of the phosphonic acid and the pK_{b} of alkyl amines, as illustrated in Fig. 1a. **c**, XPS analysis of 2PACz on ITO film. High-resolution spectra of C1s of 2PACz, HBzA, and a mixture of 2PACz+HBzA onto ITO after washing with DMF/DMSO.

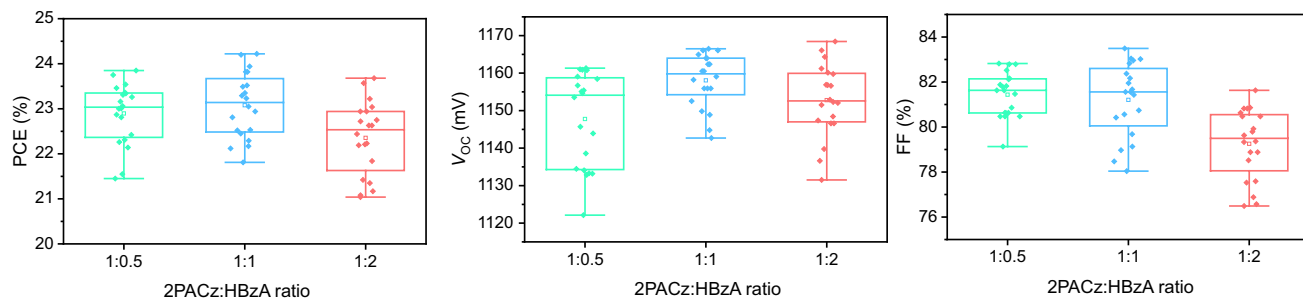


Extended Data Fig. 2 | SKPM mapping images and distributions of WF.

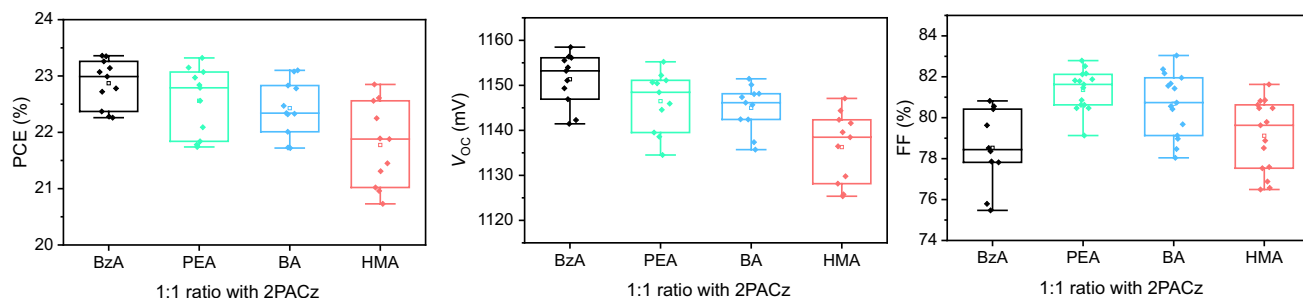
a, ITO/2PACz+HBzA sample before and after washing. The DMF/DMSO solvent mixture was used to wash the ITO/2PACz+HBzA sample. The minor WF shifting of the ITO/2PACz+HBzA sample before and after washing with DMF/DMSO indicates a strong attachment of HBzA molecules onto 2PACz molecules, consistent with XPS results. **b**, ITO/2PACz sample after washing. **c**, Bare ITO.

d, SKPM mapping images of ITO/2PACz/perovskite and ITO/2PACz+HBzA/perovskite samples after washing with DMF/DMSO. **e**, Distributions of WF of both samples from KPFM measurement. After the perovskite film deposition and subsequent removal of the HBzA ligands, the WF of ITO/2PACz+HBzA is shifted back to the WF of pristine ITO/2PACz (~5.40 eV), confirming our hypothesis.

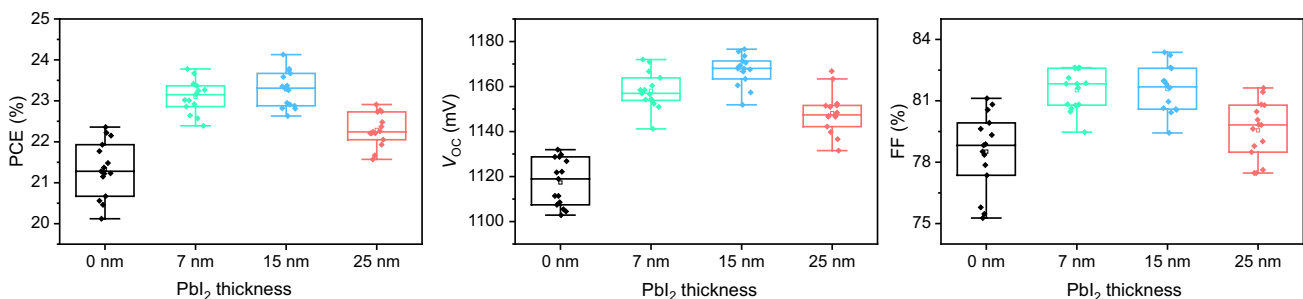
a Bottom-side 2D/3D heterojunction



b 2D/3D at bottom with various ligands

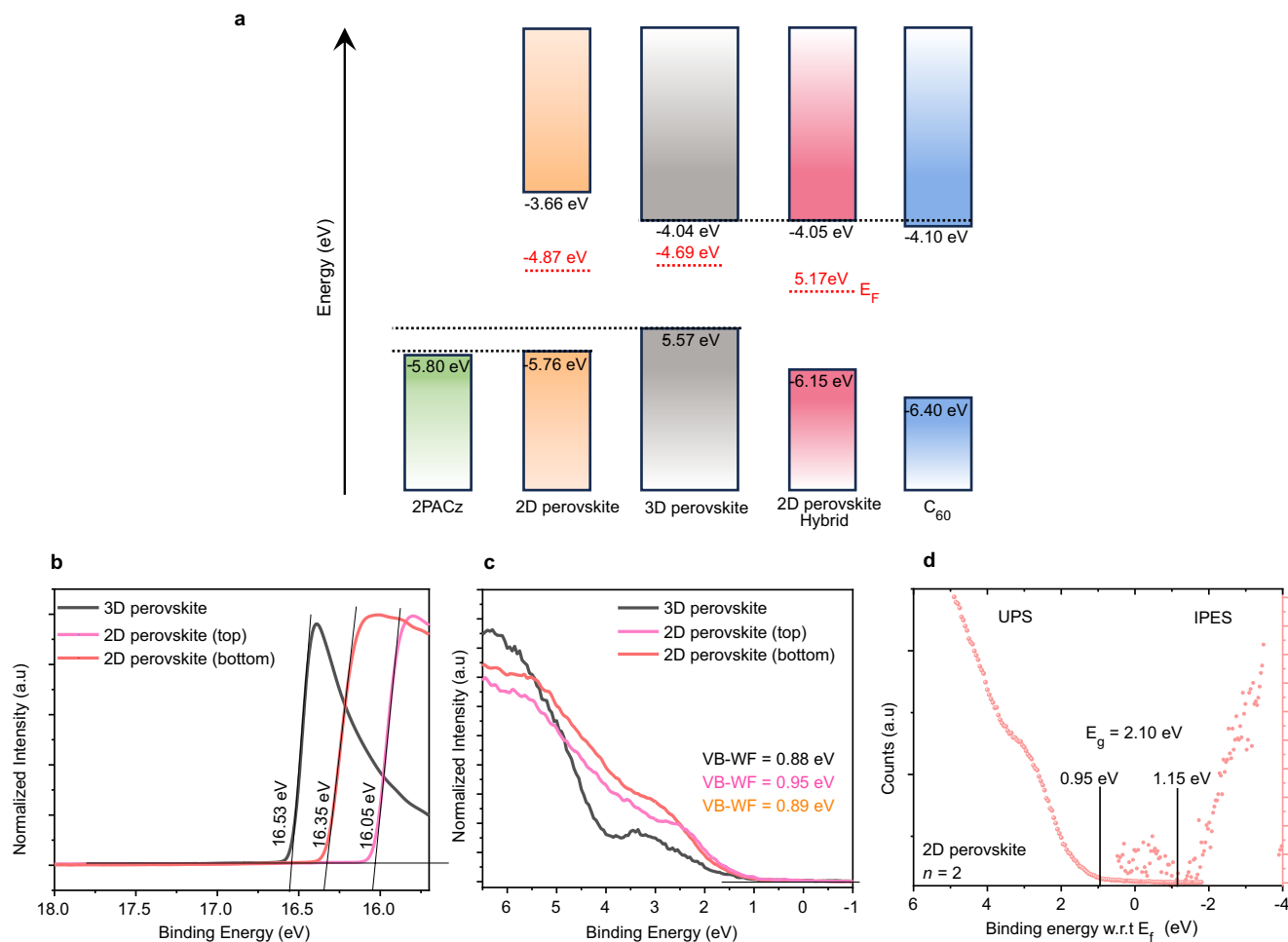


c Top-side 3D/2D heterojunction



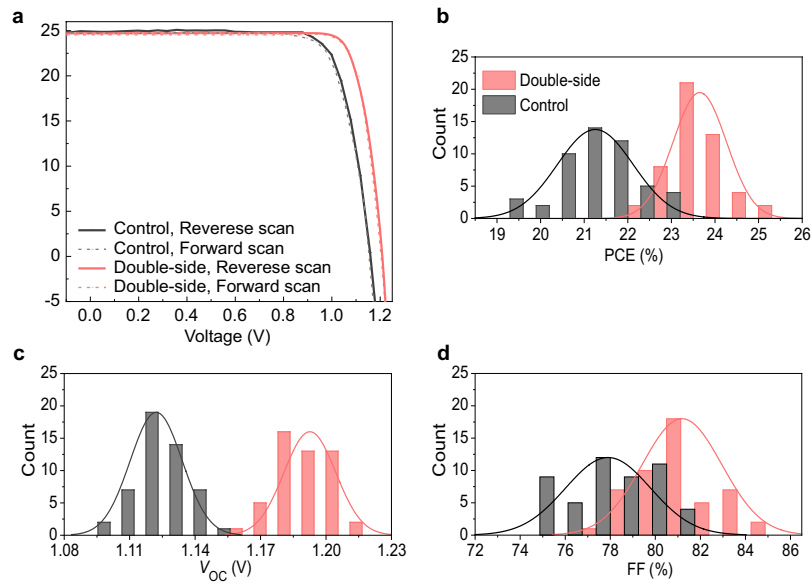
Extended Data Fig. 3 | Statistics of PCE, V_{oc} , and FF of PSCs. a, Bottom-side 2D/3D with different molar ratios of 2PACz: HBzA. **b**, Bottom-side 2D/3D with various 2D ligands (benzylamine, BzA; phenethylamine, PEA; butylamine, BA; hexylmethylamine, HMA) for bottom-side 2D/3D heterojunction with a 1:1

molar ratio of 2PACz. **c**, Top-side 3D/2D with PbI_2 thickness variation for top-side 3D/2D heterojunction. The statistics are obtained from 10 to 20 cells for each condition from different batches. It is noted that the perovskite used here is based on $CS_{0.05}FA_{0.95}PbI_3$.



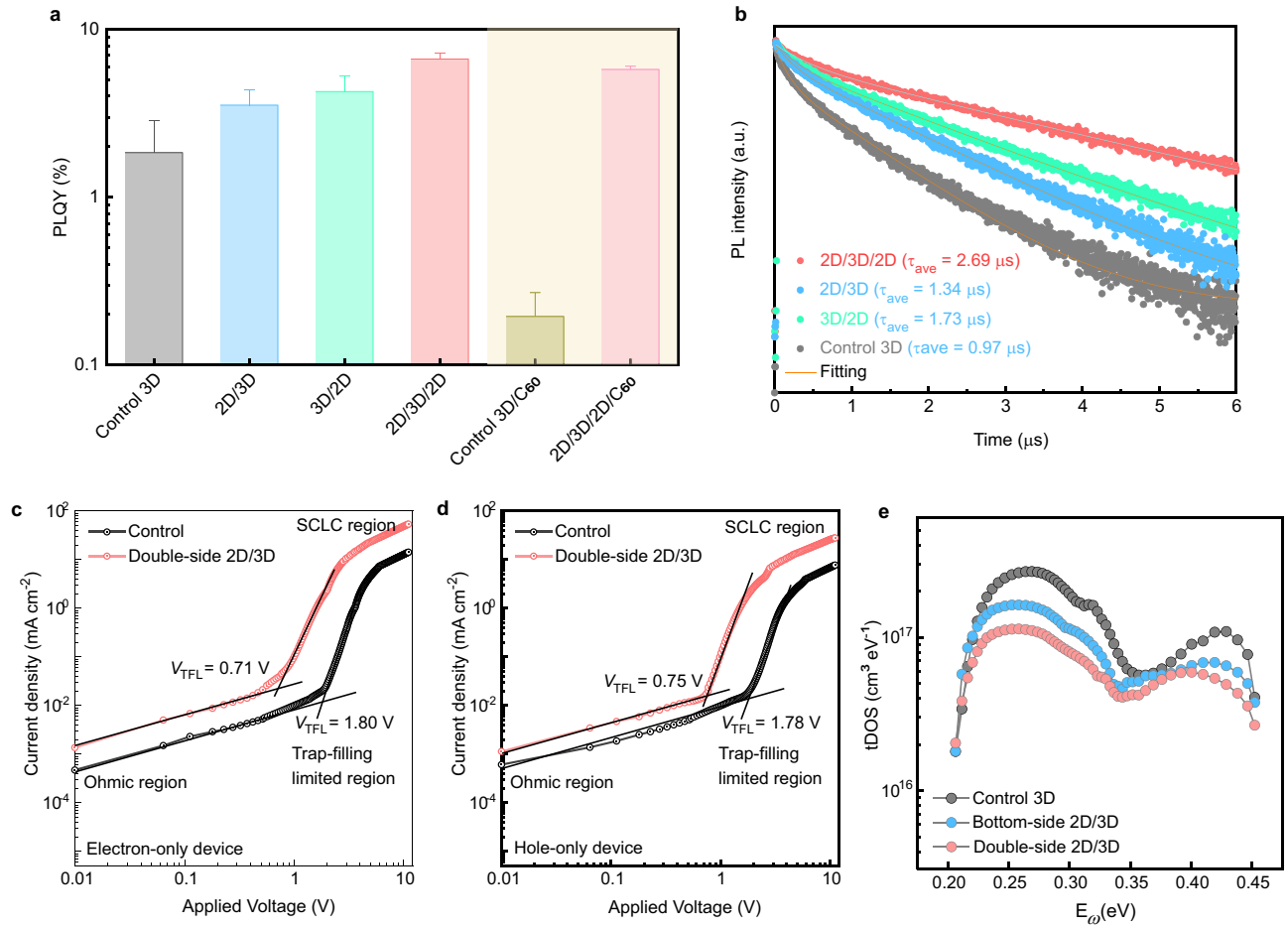
Extended Data Fig. 4 | Energetic alignment analysis. **a**, Energy level scheme for 2PACz anchored on ITO, 3D-perovskites, 2D-perovskites on top and bottom interfaces, and C_{60} layers extracted from UPS data. The VBM was obtained as $h\nu - (E_{\text{cutoff}} - E_{\text{VB,min}})$. The position of the CBM with respect to the VBM was defined

by extracting E_g (-1.53 eV) from EQE in Supplementary Fig. 8 for 3D-perovskite and by IPES for 2D-perovskite with $n = 2$ ($E_g = -2.1$ eV). **b**, **c**, Secondary electron cut-off and valence spectra plots from UPS measurement. **d**, IPES spectra of the 2D perovskite film with $n = 2$ using the hybrid method.



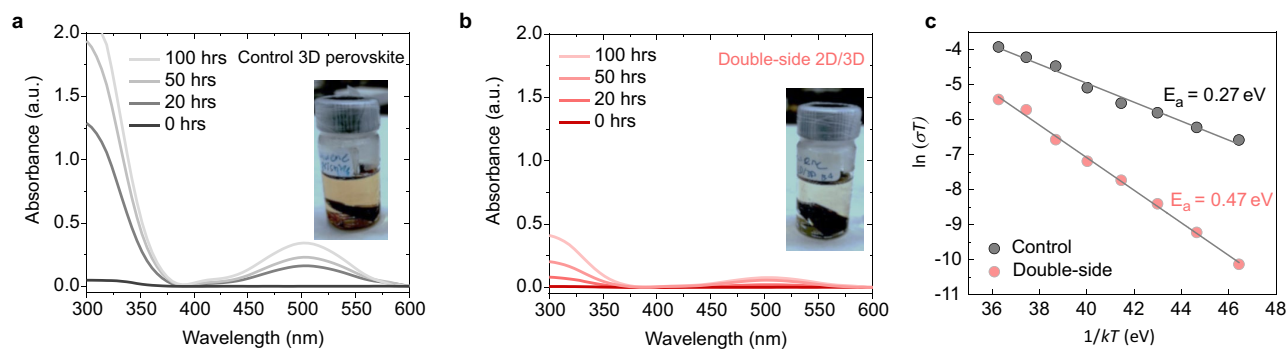
Extended Data Fig. 5 | Device performance with double-cation perovskite.
a, Representative $J-V$ curve for control and double-side 2D/3D heterojunctions perovskite solar cells with $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$ perovskite, achieving a maximum

PCE of 25.2% under reverse scan. Statistical parameters of **(b)** PCE, **(c)** V_{OC} , and **(d)** FF values of PSCs. The statistical parameters are obtained from 50 cells for each condition from different batches.



Extended Data Fig. 6 | Passivation and trap analysis. **a**, PLQY results, and **(b)** transient PL decays of perovskite films on various 2D/3D heterojunctions. **c**, **d**, SCLC plots for control and double-side 2D/3D heterojunction-based devices

of **(c)** electron-only and **(d)** hole-only devices. **e**, TAS analysis of control, bottom-side, and double-side 2D/3D passivation-based devices.



Extended Data Fig. 7 | Stability and ion migration analysis. a, b, UV-vis absorbance spectra evolution of iodine in toluene solution for the control (a) and double-side passivation of perovskite films (b) at different aging times.

Inset is a photo of the sealed vials with the control and 2D/3D perovskite films, taken after 100 h of light and heat exposures. c, Temperature-dependent conductivity of control and double-side 2D/3D perovskite films.